

Step-1

- > Create New Project
- > select Java With Ant
- > Select Java Application
- > Write the Project Name
- > Finish

Step-2

- > Open Project Window
- > Right click on Project name > New > Java Interface > Name the interface (PalindromeChecker) > Finish
- > Write code in PalindromeChecker interface

[Code]

```
import java.rmi.Remote;
import java.rmi.RemoteException;
public interface PalindromeChecker extends Remote {
    public boolean checkPalindrome(String s) throws RemoteException;
}
```

Step-3

- > Open Project Window
- > Right click on Project name > New > Java Class > Name the Class (PalindromeServer) > Finish
- > Write code in PalindromeServer class

[Code]

```
import java.rmi.registry.Registry;
import java.rmi.registry.LocateRegistry;
import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;
public class PalindromeServer extends UnicastRemoteObject implements PalindromeChecker {
    public PalindromeServer() throws RemoteException{
        super();
    }
    @Override
    public boolean checkPalindrome(String s) throws RemoteException {
```

```

        // remove all non-alphanumeric characters and convert to lowercase
        String cleanString = s.replaceAll("[^a-zA-Z0-9]", "").toLowerCase();

        // check if the string is equal to its reverse
        String reversedString = new StringBuilder(cleanString).reverse().toString();
        return cleanString.equals(reversedString);
    }

    public static void main(String[] args) {
        try {
            // create and export the server object

            // bind the object to an RMI registry
            Registry registry = LocateRegistry.createRegistry(9999);
            registry.rebind("PalindromeChecker", new PalindromeServer());
            System.out.println("PalindromeChecker server ready.");
        }

        catch (RemoteException e) {
            System.out.println("exception"+e);
        }
    }
}

```

Step-3

--> Open Project Window
 --> Right click on Project name > New > Java Class > Name the Class (PalindromeClient) > Finish
 --> Write code in PalindromeClient class

[Code]

```

import java.rmi.NotBoundException;
import java.rmi.registry.Registry;
import java.rmi.registry.LocateRegistry;
import java.rmi.RemoteException;

public class PalindromeClient {
    public static void main(String[] args) throws RemoteException{

```

```

        PalindromeClient C=new PalindromeClient();
        C.connectRemote();
    }
    public void connectRemote()throws RemoteException
    {
        try
        {
            Registry registry = LocateRegistry.getRegistry("localhost",9999);
            PalindromeChecker checker = (PalindromeChecker)
registry.lookup("PalindromeChecker");

            // test some strings
            String[] strings = {"racecar", "hello", "A man a plan a canal Panama", "12321"};
            for (String s : strings) {
                System.out.println(s + " is " + (checker.checkPalindrome(s) ? "" : "not ") + "a
palindrome.");
            }
        } catch (NotBoundException | RemoteException e) {
            System.err.println("PalindromeChecker client exception: " + e.toString());
        }
    }
}

```

Step-4

--> Clean And build all three files (PalindromeChecker, PalindromeServer, PalindromeClient)

Step-5

--> Go to project window < Select project < Libraries < Right click on libraries < Add Jar files > ()

Step-6

--> Run PalindromeServer File

Step-7

--> RunPalindromeClient file